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Jingli Li & Jianjun Zhang

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The Influence of Political Leaders' Difference on Government Leader's career development in China: The Role of Capability, Seniority and Relationship

ABSTRACT: This study analyses 1,870 pairs of party secretaries and mayors in prefecture-level cities from 1993 to 2020 to examine how leader team configurations affect mayors' career development. It focuses on the roles of seniority, capability, and relationships between leaders. Results show that mayors' career advancement is more likely when the party secretary has greater seniority—such as longer party tenure, earlier office assumption, or local origin—especially if the mayor is from another city. Comparable capabilities between the two leaders reduce the mayor's career prospects, while differences in their relationship have non-significant impact. This research contributes empirical evidence highlighting the importance of seniority and capability in leader team configurations, suggesting political parties should consider these factors to promote effective governance.

KEYWORDS capability, seniority, relationship, government leader's career development

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1.Introduction

Presently, most studies focusing on the career development of party and government cadres tend to adopt an individual-centric approach, concentrating on factors such as individual abilities and performance (Bo, 2002; Crossland et al., 2014; Li and Zhou, 2005), relationship networks (Choi, 2012; Shih et al., 2012; Opper et al., 2015), and their consequential impact on career progression (Jia et al., 2015; Luo and Qin, 2021; Zeng and Zhou, 2024). However, there exists a noticeable void in the discussion concerning the influence of the pairing of party and government leader teams on the career development of cadres. This gap is regrettable because the synergy between the party secretary and the executive officer, along with the alignment of party and government

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leadership teams, not only impacts organizational performance but also significantly shapes the career trajectory of its leaders, particularly those in government roles.

Rooted in traditional Chinese Confucian and Legalist philosophies, the pursuit of order holds paramount importance in governance, with a natural inclination toward hierarchy fostering stability (Chen and Lee, 2008). Consequently, the concept that relative centralization within top management teams can yield order, stability, and positively impact performance possesses distinct Chinese characteristics and relevance. The observable positive outcomes of team hierarchy disparities further substantiate this point (Xie et al., 2020).

However, it is essential to recognize that a leadership team with a capable leader, but inept subordinates may impede scientific decision-making and hinder the mobilization of subordinates' initiative. Likewise, a party and government team characterized by a robust core and feeble branches is detrimental to economic and social development, cadre training, and the sustained progress of the party and government leadership team.

In China's political system, the Communist Party exercises unified leadership over both party and government officials. At the local level, the mayor serves as the head of the municipal government, while the municipal party secretary is the top leader in the city, wielding superior political authority. The career development of mayors is not determined by public elections but is managed through the hierarchical cadre management system under the leadership of the Chinese Communist Party (CCP).

The selection, promotion, and demotion of mayors are primarily controlled by the Organization Department of the CCP at the provincial level, which oversees personnel arrangements in prefecture-level cities. The municipal party secretary plays a critical role in this process. As the top official in the locality, the party secretary evaluates the mayor's performance,

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provides political assessments, and often makes recommendations regarding the mayor's future appointments. These recommendations are crucial inputs into the decision-making process of the provincial party committee and the Organization Department.

Mayoral appointments typically follow two career paths: promotion to higher-level administrative positions (e.g., vice-governor, provincial-level party secretary), lateral transfer to equivalent posts in other cities, or transition into party leadership roles within the same locality. Conversely, career stagnation or demotion can occur when mayors are transferred to non-executive roles, removed from office, or assigned to positions such as chair of the local People's Congress or CPPCC, especially before retirement age.

This politically driven appointment and assessment system creates unique dynamics within Chinese local governance. The relationship between the party secretary and the mayor becomes central to both governance performance and individual career trajectories. Factors such as seniority, competence, and relational networks play crucial roles in determining whether the mayor is supported or sidelined by the party secretary. The balance of power between mayors and party secretaries in China is influenced not only by professional expertise, knowledge, and experience, but also by relational resources such as networks and political connections. In addition to competence and seniority, relational resources play a pivotal role in shaping the dynamics between party secretaries and mayors as well as their subsequent career outcomes. While competence reflects leaders' ability to deliver performance and seniority signals accumulated institutional legitimacy, relational resources provide access to political sponsorship, protection, and opportunities that may not be attainable through merit alone. Leaders with strong relational capital can offset limitations in professional competence or relatively short tenure, thereby maintaining influence and improving their chances of upward mobility. Conversely,

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even highly competent or senior leaders may encounter career stagnation or demotion if they lack critical political connections or suffer from strained relationships with their counterparts. This suggests that career trajectories in the Chinese political context are not determined solely by individual merit or experience but emerge from the interplay of competence, seniority, and relational resources, with relationships often serving as the decisive factor in resolving power imbalances within leadership pairings.

This study scrutinizes the impact of variations in seniority, competence, and relationships among party and government leaders on the career development of government leaders. Adhering to the "party-controlled cadres" principle, where the mayor's authority is subordinate to that of the municipal party secretary, any discord or conflict between them disproportionately affects the mayor. Given that mayors often ascend to become the secretary of the municipal party committee in the future, this research holds particular significance for successor development. The study underscores that a good-matched pairing of party and government leaders significantly heightens the odds of government leader's career development. However, limited research exists on this crucial topic, emphasizing the necessity to delve into the matching dynamics of party and government leaders.

Focusing on the party and government leaders of prefecture-level cities, this research posits that a party secretary with higher seniority not only aligns with tradition and meets subordinates' expectations, thereby fostering team order, but also proves beneficial for individual career development. In terms of capability, an equitable blend of strength between the two leaders complies with cadre selection standards and enhances team performance, consequently increasing the likelihood of the mayor's career development. Regarding relationships, we contend that a secretary with a more extensive network contributes to maintaining team

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harmony, reducing the risk of conflicts, and subsequently increasing the chances of the mayor's career development.

Following an analysis of 1870 sets of paired data encompassing the experiences and characteristics of municipal party secretaries and mayors in each city from 1993 to 2020, this paper substantiates the propositions to a certain extent.

This study offers valuable contributions on multiple fronts. Firstly, it diverges from the predominant focus on the individual characteristics of party and government cadres and their impact on career development, instead pioneering an investigation into the influence of party and government pairing on individual career development. This novel perspective enriches our understanding of the career trajectories of leadership cadres. Secondly, the paper meticulously examines the effects of various pairings within party and government leader teams on mayoral career development, considering differences in seniority, capability, and relationships. This groundwork provides a foundation for optimizing the configuration of party and government leader teams. Moreover, by introducing the crucial concept of seniority and empirically discussing its role in maintaining internal order within party and government leader teams, this research lays the groundwork for future exploration.

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2.Theory and Hypotheses

There have been relatively few studies conducted on the collocation of party and government leader teams, despite some theoretical reviews. Existing literature predominantly concentrates on studies of individual cadres, often emphasizing personal characteristics and relationships as influential factors in career development. In empirical studies, the competence of party and government cadres is commonly evaluated based on their career experiences and education levels. This paper endeavors to bridge this research gap surrounding party and

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government leader teams. Specifically, it posits that various combinations of party and government leaders may yield different impacts on their collaboration, and conflicts within these teams could adversely affect their career trajectories. Furthermore, the influence of party and government leader teams on regional performance can, in turn, impact the career development of individual cadres.

Moreover, this paper contends that, in the Chinese context, seniority is an essential dimension that cannot be overlooked. While competence and relationships are frequently discussed, differences in seniority among party and government leaders can play a crucial role in maintaining team cohesion. Seniority becomes a distinguishing factor when the competence of two leaders is comparable. Consequently, the ensuing section will delve into the specific effects of disparities in seniority, competence, and relationship among party and government leaders on a mayor's career development.

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Seniority: Impact of differences in seniority between party and government leaders on mayor's career development

Seniority in China is a concept deeply ingrained in the cultural fabric, reflecting distinctive Chinese characteristics and rooted traditions. This notion finds its roots in Confucian philosophy, a profound influence on Chinese culture, which underscores the importance of respecting elders and upholding hierarchical relationships. Traditional Chinese workplaces further exemplify the emphasis on seniority, where deference to experienced individuals is paramount, translating seniority into authority and decision-making power within a hierarchical structure. Historical practices, notably the civil service examination system, dating back to imperial China, selected officials based on both knowledge and demonstrated experience, reinforcing the tradition of valuing seniority. Linguistically, the Chinese language contains expressions like "论资排辈"

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(lùn zī pái bèi), directly translating to “ranking based on qualifications and seniority”, highlighting the intertwined significance of expertise and experience. This cultural emphasis on respect for elders extends across various aspects of life, including family, education, and the workplace. The concept contributes to social harmony, providing a stable and predictable structure in interpersonal relationships and organizations, essential for maintaining order and cohesion within groups. In essence, the claim that seniority is a concept with distinctive Chinese characteristics is substantiated by the multifaceted interplay of cultural, historical, linguistic, and social factors that prioritize qualifications and experience.

Research on seniority has traditionally centered on its implications for organizational management and its correlation with wages, as demonstrated in studies (Claessen, 2023; Geys et al., 2020). While this perspective has garnered significant attention, some scholars have delved into the specific importance of seniority within the context of Chinese culture, particularly concerning multinational corporations operating in China. Fu and Kamenou (2011) have identified seniority as one of the three primary cultural values shaping human resource management (HRM) practices in China, alongside relationship dynamics and people-oriented values. The salience of seniority in the Chinese workplace aligns with the prevalent high power-distance, where subordinates typically accept power distribution based on hierarchical positions.

Seniority holds deep roots in Chinese culture, stretching back thousands of years to its origins in early Chinese ritual order and later formalized by Confucianism. This enduring cultural trait plays a pivotal role in shaping organizational dynamics and interpersonal relationships. Similarly, in Japan, the significance of seniority is reflected in fundamental components of traditional HRM, such as seniority-based pay, promotion systems, and the existence of corporate unions, where age and seniority serve as visible markers of hierarchical

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status (Sano, 1995). However, the landscape of corporate HRM practices in Japan has undergone a transformation, gradually shifting towards performance-based pay and promotion mechanisms in response to the intensifying demands of the competitive business environment (Zhu et al., 2007).

Seniority also holds a pivotal role in public management, particularly within the party and government leader teams in China, as argued by Zang (2001). Zang contends that party seniority is indicative of political reliability and should not be overlooked in the assessment and promotion of officials. Additionally, party standing serves as a tangible measure of seniority. In a study comparing the party standing of provincial party and government leaders in 1988, Zang discovered that provincial party secretaries exhibited longer party standing (38.1 years) compared to government leaders (35.1 years). Within this framework, seniority is suggestive of accumulated experience, expertise, and elevated prestige, conferring authority upon cadres. In the Chinese political context, the higher seniority of party secretaries proves advantageous in upholding authority, fostering order within the team, mitigating conflicts, promoting team harmony, and ultimately benefiting the career development of cadres. Consequently, mayors are more likely to ascend to the position of local municipal party secretaries, ensuring that secretaries commonly possess higher seniority in a given locality than mayors.

Building on this perspective, Haleblian and Finkelstein (1993) contend that an extended tenure generally corresponds with augmented power, providing executives with opportunities to amass social capital, fortify their positions, and consolidate their power bases (Greve and Mitsuhashi, 2007). Similarly, insights from CEO succession research suggest that CEOs recruited from outside the firm, referred to as parachutists, face an increased likelihood of turnover due to their limited social network resources within the organization (e.g., Shen and

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Cannella, 2002). In essence, if the municipal party secretary boasts higher seniority, such as possessing longer party standing than the mayor or assuming a leadership role, the resulting team structure exhibits a more pronounced order, and the probability of conflict is substantially reduced. Consequently, this increases the likelihood of the mayor's career development.

Based on the above, we propose the following hypotheses:

Hypothesis 1*: If the municipal party secretary has longer party standing than the mayor, the mayor has a higher chance of career development.*

Hypothesis 2*: If the municipal party secretary takes office earlier than the mayor in the same city, the mayor has a higher chance of career development.*

Hypothesis 3*: If the municipal party secretary is local and the mayor is transferred from outside, the mayor has a higher chance of career development.*

Competence: Impact of differences in the competence of party and government leaders on the career development of mayor

In contrast to the explicit nature of seniority, the competence of party and government leaders encompasses a more implicit variable, typically represented by functional background diversity, education, and political performance. The diversity in functional backgrounds serves as a proxy for the expansive range of experiences and depth of knowledge among leaders, facilitating more effective collaboration and integration across departments and individuals to address uncertainties. Moreover, diverse functional backgrounds contribute to the cultivation of a holistic perspective and the generation of diverse solutions. Empirical evidence substantiates the positive impact of functional diversity. For instance, Crossland et al. (2014) discovered a positive influence of CEO work experience diversity on the novelty of firm-level strategy among CEOs in the top 250 Fortune companies. Similarly, Hitt and Tyler (1991) argued that executives with

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varied functional backgrounds are better equipped to establish comprehensive criteria in corporate merger negotiations. Rajagopalan and Datta (1996) concluded that the diversity in functional backgrounds enhances the breadth of knowledge and skills, facilitating a multidimensional diagnosis of business problems.

In the realm of public management, work experience in diverse sectors tends to nurture varied perspectives, knowledge, abilities, and relationships. Party leaders with multifaceted work experiences are likely to exhibit higher levels of work competency. Education, often linked to cognitive ability, attitudes, and expertise (Bantel and Jackson, 1989; Knight et al., 1999; Wiersema and Bantel, 1992), has demonstrated its significance in organizational innovation, as observed in studies on hospitals (Kimberly and Evanisko, 1981) and banks (Bantel and Jackson, 1989). Moreover, investigations into the career development of party and government leaders indicate that officials with higher educational levels are more apt to formulate timely and effective policies during emergencies (Li et al., 2022) and progress in their careers (Chen et al., 2019).

The synergy between a competent party secretary and mayor contributes to enhanced team performance. Conversely, a substantial gap between the municipal party secretary and the mayor in terms of functional background diversity or educational level elevates the likelihood of the mayor’s career development. This stems from the potential inadequacy of the mayor in performing the role effectively when capabilities are low. Conversely, if the mayor's capabilities surpass those of the party secretary, it may pose a challenge to the authority of the municipal party secretary. Therefore, when the functional background diversity and education of the party secretary and mayor are roughly equivalent, the probability of the mayor’s career development increases.

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Based on the above, we propose the following hypotheses:

Hypothesis 4*: If the education level of the party secretary and the mayor is similar, the chances of the mayor's career development would be higher.*

Hypothesis 5*: If the functional background diversity of the party secretary and the mayor is similar, the chances of the mayor's career development would be higher.*

Relationships: Impact of differences in party secretary and mayor's relationships with provincial leaders, and the impact of differences in party secretary and mayor's powerful sector experience on mayor's career development

The significance of relationship networks in Chinese society is widely acknowledged, with numerous studies investigating their impact on officials' career trajectories. Notably, extensive research has delved into the influence of informal relationships between officials and their supervisors on promotion prospects. Choi (2012), for instance, emphasized the critical role of personal relationships with key leaders, factional ties, and economic performance in the career success of provincial officials. Leutert (2018) highlighted the substantial impact of political ties and patronage-type relationships on promotion opportunities. In a detailed observation of cadres in a county-level city, Feng (2010) identified the significance of officials being classmates or hailing from the same hometown in cadre selection. Chen (2019) argued that serving as a leader's secretary could be considered part of the social network, establishing a positive correlation between secretary experience and promotion.

However, existing studies have not sufficiently differentiated between party and government leaders. Zuo (2015) asserted that the incentives for promotion differ between local municipal party secretaries and mayors. Specifically, personal connections with provincial

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leaders and strong economic performance are deemed prerequisites for the promotion of party secretaries, while mayors' age, the city's level, and the effectiveness of social policies become critical factors in their career advancement.

Regarding the relationships between party and government leaders and their superiors, three plausible scenarios emerge. First, the party secretary may possess more relationships than the mayor, fostering an environment where the mayor acknowledges the party secretary's authority to maintain internal order and reduce the likelihood of conflict, thereby facilitating the career development of both officials. Second, if the mayor has more relationships than the party secretary, this could propel the mayor's career advancement but may also lead to arrogance and a challenge to the party secretary's authority, resulting in a lose-lose situation. The impact of relationships on the mayor's career development becomes unclear in this case. Third, if the relationships with provincial leaders between the two officials are equal, their interactions may vary, leading to either competition or cooperation, thereby rendering the mayor's career development uncertain.

Based on these considerations, we propose the following hypotheses:

Hypothesis 6*: If the municipal party secretary has more hometown relationships with provincial party leaders than the mayor, the chance of career development for mayor is higher.*

Hypothesis 7*: If the municipal party secretary has more alumni ties with provincial leaders than the mayor, the chance of career development for mayor is higher.*

Hypothesis 8*: If the municipal party secretary has more experience in the power sector (Communist Youth League, Organization Department, or as a secretary) than the mayor, the chance of career development for mayor is higher.*

3. Sample and methodology

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3.1 Sample

Our sample comprises 333 prefecture-level administrative regions and encompasses both party secretaries (referred to as "secretaries") and mayors. We gathered information on secretaries and mayors spanning from 1993 to 2020, utilizing publicly available official sources such as the People's Website Local Leaders Database, Economic Website Local Party Leaders Database, official websites of each region, and Baidu Encyclopaedia. Demographic information and work experience of secretaries and mayors, including gender, ethnicity, age, party standing, duration as mayor, location of the last position, departure time of the last position, next position, education level, and functional experiences, were manually collected. Economic data, such as whether the regions held the status of important cities (sub-provincial) and GDP growth rates by year, were obtained from the Chinese Economy Statistical Database.

Consistent with established research practices (Zuo, 2015), we excluded data from Beijing, Tianjin, Chongqing, and Shanghai due to their distinct status as municipalities directly under the central government, leading to variations in the selection and appointment processes of their leaders. Paired data were constructed by merging secretary and mayor information using Stata, and samples were constructed based on time. Samples with significant missing information, instances where one leader served as both secretary and mayor, and cases involving retirements, resignations, and deaths were systematically excluded.

3.2 Variable

3.2.1 Dependent variable

We are examining the combined impact of the secretary and mayor pairing on the career development of the mayor. Therefore, the dependent variable in our study indicates whether the

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subsequent position of the mayor constitutes career development (promotion or lateral transfer), coded as 1, while any other changes, such as demotion, is coded as 0. Although lateral transfer may not immediately signal upward mobility, it can nonetheless be positively related to long-term career development. In the Chinese political system, leaders who are transferred laterally across different cities accumulate diverse governance experience, broaden their administrative horizons, and demonstrate adaptability to varying local conditions. Such cross-regional assignments are often interpreted as a test of capability, as they require leaders to handle heterogeneous economic, social, and political environments. The breadth of experience gained through multiple city postings not only enhances leaders' professional competence but also signals their potential for higher-level responsibilities. Consequently, lateral transfers should be viewed as an important mechanism for capability development and as a pathway that increases the likelihood of subsequent career advancement.

Our demotion definition aligns with established research practices, as seen in works like Zuo (2015). It encompasses scenarios where a mayor is appointed to an ordinary city from a sub-provincial city or takes on a director position of the People's Congress or chairman of the CPPCC within the same city, provided the mayor has not reached retirement age.

Additionally, we considered cases where cadres were initially promoted but subsequently demoted due to prior issues, classifying their subsequent positions as demotions. To ensure the precision of cadre career development classification, we implemented a delay in observing the subsequent positions of transferred cadres. For instance, a mayor who retired after serving as the director of the municipal People's Congress was classified as undergoing a lateral transfer.

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3.2.2 Independent variable

The independent variables in our analysis encompass differences in seniority, competence, and relationships between the secretary and mayor. Seniority is gauged by party standing, the date of appointment as mayor, and whether the official was appointed locally. Party standing is quantified in years, calculated from the day the official joined the party to the day the secretary and mayor commenced their collaboration. The date of appointment as mayor denotes the specific day the official assumed the role. Officials appointed from within the local city are categorized as locally appointed. In our analysis, we concentrate on the disparities between the secretary and the mayor in these dimensions. Therefore, we consider the secretary having longer party standing, an earlier appointment, and a local appointment compared to the mayor as indicative of higher seniority for the secretary, coded as 1, with other cases coded as 0.

Competence is proxied by education level and functional experience diversity. Education is widely regarded as an indicator of leaders' cognitive ability, formal training, and analytical skills, all of which are important for effective governance. In the Chinese political system, higher educational attainment also signals professional credibility and legitimacy. Functional experience diversity, in turn, reflects the breadth of roles and responsibilities leaders have undertaken across different departments, regions, or sectors. Leaders with more diverse career paths are assumed to possess broader organizational knowledge, adaptive problem-solving capacity, and cross-sector coordination skills. Together, these two proxies capture both the knowledge foundation and the practical skill set and adaptability that are essential dimensions of competence. However, these proxies cannot fully reflect actual managerial or political abilities, and their use may bias results toward observable characteristics while underestimating unmeasured aspects of competence. Possible alternative measures for future research, such as performance evaluations, leadership

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assessment scores, or longitudinal indicators of policy achievements, should be adopted if the data is available.

Education level is classified as high school and below (coded as 1), college (coded as 2), bachelor's (coded as 3), master's (coded as 4), and doctorate (coded as 5). Functional experience diversity is determined by the number of experiences in various sectors, including economic, judicial, organizational, health, and environmental sectors, as well as work experience in universities, enterprises, and as secretaries of leaders. An official with experience in any of these sectors is coded as 1. The similarity in competence between the secretary and mayor is quantified by the absolute difference in their functional experience diversity and education level. Smaller absolute values indicate more similar levels of competence.

Relationships between the secretary and mayor are gauged by shared hometown or alma mater, and experience in powerful departments (such as the Communist Youth League, the Organization Department, or as secretaries of higher-level leaders). "Same hometown" is determined by whether the secretary or mayor's household registration province matches that of the provincial party secretary and governor. "Alumni" is determined by whether the secretary or mayor attended the same university as the provincial party secretary and governor. If the secretary has more relationships than the mayor, including more hometown and alma mater connections and more experience in powerful departments, it is coded as 1; otherwise, it is coded as 0.

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3.2.3 Control variable

Incorporating the well-established finding from previous literature that local economic development significantly influences officials' promotion prospects (e.g., Li and Zhou, 2005), we introduced the average GDP growth rate of the city during the mayor's term as a control

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variable (Bo, 1996). Additionally, we accounted for factors that might impact career prospects, such as officials' age, gender, ethnicity, education level, and promotion speed (Landry, 2008; Walder, 1995). Gender was coded as 1 for female and 0 for male, ethnic minorities as 1 and others as 0, while the reciprocal of officials' tenure as mayors served as a proxy for promotion speed. Recognizing that sub-provincial cities hold higher status, and mayors of such cities have an increased chance of promotion (Zuo, 2015), we coded sub-provincial cities as 1 and ordinary cities as 0. Additionally, Regional variables are coded as categorical indicators representing the different geographic areas in which leaders served. These categories are used to control for regional variation in political and economic contexts, without assuming any inherent order among them.

Finally, considering the substantial changes in cadres' performance appraisal in 2007 (Jiang et al., 2015), we designated 2007 as a reference year and coded the situation before and after the change as 0 and 1, respectively.

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3.3 Empirical results

A binary logistic model was used for the analysis, given that the dependent variable is dichotomous.

3.3.1 Model goodness-of-fit test

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Table 1 shows that the Hosmer and Lemeshow test yielded a significance level greater than 0.05, indicating a good fit of the model.

3.3.2 Descriptive Statistics

Among the 1900 mayors in our sample, 1749 individuals (92.1%) experienced career development, either through promotion or lateral transfer to an equivalent position or institution with similar administrative rank and power.

Regarding seniority, in 1175 cases (61.8%) the party secretary had a longer party standing than the mayor, while in 923 cases (48.6%) the party secretary assumed office earlier than the mayor in the same city.

For local appointment status, 356 party secretaries (18.7%) were promoted from within the same locality, while the remaining were assigned from other regions.

In terms of relational factors, 227 pairs (11.9%) shared a schoolmate or alumni relationship, and 684 party secretaries (36.0%) had previous experience in powerful central or provincial departments, which could potentially influence their political network and resource access.

3.3.3 Empirical findings

The results of the binary logistic regression analysis are presented in Table 2. Hypothesis 4, which posited that the mayor has higher chance of career development when the secretary and mayor have comparable education levels, was not supported. However, when we used whether the mayor's education was higher than that of the secretary as a proxy, we found that the mayor's

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chance of career development increased by about 45%. This suggests that mayors with higher education, particularly higher than that of their cooperative secretaries, have a greater likelihood of being promoted, highlighting the advantage of intellectualization in cadre promotion.

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Hypotheses 6, 7, and 8, which stated that the mayor has more chance of career development when the secretary has more relationships with provincial leaders than the mayor, were not significant. The reasons behind this finding are worth considering. One possible explanation is that higher authorities may be more willing to break “subtle rules” and prioritize the promotion of cadres at the grassroots level, as well as strengthen the rotation and exchange of cadres. As a result, officials from other provinces may be transferred to the province and cadres at the grassroots level may be promoted. This continuous improvement of the political ecology may make a cadre’s comprehensive capability a more important factor in selection than relationships.

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The remaining hypotheses were all confirmed. Specifically, regarding seniority differences, secretaries with longer party standing than mayors accounted for approximately 62% of the total sample. When secretaries had longer party standing, the odds of career development for mayor were about 32% higher than in other cases. Additionally, secretaries who took office earlier than mayors accounted for half of the total sample, and when this was the case, the odds of career development for mayor were about 40% higher than in other cases. Furthermore, the sample of secretaries who were appointed locally while the mayor was transferred from another region accounted for about 19% of the total sample. In this case, the odds of career development for mayor were about 43% higher than in other cases, thus verifying hypotheses 1-3. In terms of capability differences, hypothesis 5 suggests that the odds of a mayor's career development will increase when the functional background diversity of the secretary and the mayor are comparable, and this was also confirmed. As shown in Table 2, when the absolute value of the functional background diversity difference between the secretary and the mayor increased by one, the odds of mayoral career development decreased by approximately 20%.

Among the control variables, the odds of a mayor's career development were lower in eastern and central cities relative to western regions, and lower in sub-provincial cities. This may be due to better economic conditions and a more important city status, which may bring about more competitive resources, causing the secretary and mayor to have inconsistent views on work and triggering discord during cooperation. Regarding GDP growth rate, the higher the average GDP growth rate, the more likely the mayor's career is to be promoted, which is consistent with the idea of promotion tournaments (Li and Zhou, 2005). With regards to the characteristics of the mayor, when the mayor is older, the chance of career development is lower, which may be a transitional procedure before retirement. Meanwhile, the higher the mayor's education level, the

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higher the chance of career development will be, confirming the importance of intellectualization in cadre appointments. After the change in the concept of performance appraisal, the odds of a mayor's career development decreased, which may be the result of the higher authority's increased focus on environmental performance and anti-corruption initiatives.

Additionally, it is necessary to acknowledge the potential endogeneity of leader pairing, which may arise if unobserved factors simultaneously influence both leader assignment and career outcomes.

3.3.4 Robustness test

We conducted four tests to assess the robustness of our model. Firstly, given that the loss of order within a team often leads to infighting, and fighting is injurious to all, and the demotion of the secretary is placed as the independent variable in the above model. Similarly, demotion is coded as 1 and other cases as 0. The results were consistent with Table 2, where the odds of mayoral career development increased by about 63% compared to other cases. This test confirmed the importance of cadre matching in party and government leader teams.

Secondly, we tested whether the differences in relationships significantly affected the mayor's development by coding mayors with more relationships and powerful department experiences as 1 and others as 0. The results were consistent with those in Table 2.

Thirdly, we coded cases where the numbers of relationships or powerful department experiences were equal as 1 and others as 0. We found that none of the three types of relationship or experience equivalence were significantly related to the mayor's career development, which validated the robustness of the model.

Fourthly, we conducted an additional robustness check by replacing the binary variable for seniority difference with a continuous measureto further validate our findings. Specifically,

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we calculated the actual gap in party standing between the party secretary and the mayor in terms of months. This approach allows for a finer-grained analysis of seniority differences. The results from this robustness test are consistent with those of the original model, confirming that our conclusions hold regardless of whether seniority is measured as a binary indicator or as a continuous variable.

4.Discussion and conclusion

This study examines how differences within municipal leadership teams are associated with the career development of mayors, focusing on differences in seniority, competence, and relational characteristics between party secretaries and mayors. Using a dataset of 1,870 matched party secretary–mayor pairs from 1993 to 2020, the empirical results suggest that the likelihood of mayoral career advancement is relatively higher when the party secretary holds greater seniority than the mayor, and when the secretary and mayor share similar functional backgrounds and educational levels. These findings provide empirical support for the proposed hypotheses, although they should be interpreted as associations rather than definitive causal relationships.

This study contributes to the literature on political leaders' career development by examining how the composition of paired party–government leadership may relate to mayoral career outcomes in China. By focusing on differences in seniority and competence between party secretaries and mayors, the research adds a team-level perspective to existing studies of cadre promotion, while situating the analysis within the institutional and cultural context of the Chinese political system.

In China's cadre management system, elements of meritocratic governance coexist with longstanding institutional and cultural traditions. While reforms have increasingly emphasized education, professional competence, and performance evaluation, hierarchical norms and

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seniority continue to shape organizational interactions and career trajectories. Seniority can influence perceptions of authority and role expectations within local leadership teams, which may in turn affect patterns of cooperation and coordination between party secretaries and mayors.

Consistent with this institutional context, the empirical results indicate that when the party secretary has greater seniority than the mayor, the mayor's likelihood of career advancement tends to be higher. One possible interpretation is that clearer hierarchical ordering within leadership teams may reduce role ambiguity and facilitate coordination. At the same time, the positive association between similarity in functional backgrounds and educational levels suggests that shared professional experiences may also contribute to smoother working relationships and more effective collaboration within leadership teams.

We recommend that higher-level organizational departments consider both seniority and competence matching when appointing party and government leader teams to maintain organizational harmony while promoting capable leadership. It emphasizes that when appointing party and government leader teams, balancing seniority and competence can simultaneously promote capable leadership and maintain functional coordination within the organization. In other words, organizational harmony is framed as a facilitator, rather than a substitute, for effective governance and high performance.

This study has several notable contributions. Firstly, it provides novel insights into the career development of party and government cadres, which is influenced not only by personal factors but also by the party and government leader team, such as seniority and ability differences. While previous studies have mostly focused on the influence of individual characteristics of party and government cadres on their career development (e.g., Bo, 2002;

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Crossland et al., 2014; Li and Zhou, 2005, Choi, 2012; Shih et al., 2012; Opper et al., 2015), our study is among the first to investigate the impact of the leader team on the career development of government leaders.

Secondly, we specifically discuss the impact of the leader team on a mayor's career development from the perspective of seniority, capability, and relationship. Previous studies have mostly focused on the capability (Chen, 2019), relationship (Choi, 2012; Shih et al., 2012), and political performance (Li and Zhou, 2005) of political leaders to explain their promotion. In contrast, our study introduces the concept of seniority, which has distinct cultural characteristics in China, and analyzes the impact of seniority differences between the secretary and mayor on the mayor's career development. This opens new avenues for future research.

Based on the findings of this study, we recommend that higher-level organizational departments consider the pairing of party and government leader teams based on individual capability when selecting and appointing cadres. Specifically, attention should be paid to three factors: seniority, capability, and relationship. Firstly, differentiating the seniority of party and government cadres is important to maintain order within the team and preserve the party secretary's authority. Secondly, it is essential to appoint cadres with strong and complementary capabilities, with multi-functional department experience and cross-regional exchange experience being conducive to their growth. High-quality education also enhances the comprehensive quality of leading cadres. Thirdly, a sound "joint meeting system" should be established to avoid the lack of cooperation between party and government leaders. It is also crucial to make important decisions, such as appointing and dismissing key cadres, arranging important projects, and using large amounts of funds, through collective discussion.

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5.Limitations

However, several limitations should be acknowledged. First, this study focuses primarily on the independent effects of seniority, competence, and relational similarities, while the potential interactive effects among these factors remain insufficiently explored. Future research could examine whether combinations of these characteristics jointly influence political career trajectories.

Second, the analysis is constrained by data availability. Although the dataset covers a large number of municipal leadership pairs, some potentially important relationships—such as connections between municipal leaders and higher-level provincial leaders—could not be systematically captured. The omission of such variables may limit the comprehensiveness of the analysis.

Third, potential endogeneity remains a concern. The pairing of party secretaries and mayors is not randomly assigned in practice. Leadership appointments may be influenced by a variety of political and administrative considerations, such as existing governance challenges, factional dynamics, or prior working relationships. These factors may simultaneously affect both leader pairing patterns and subsequent career outcomes. Due to the lack of detailed appointment process data, this study is unable to fully address this issue. Future research could employ more refined identification strategies or richer datasets to better assess the causal mechanisms underlying leadership pairing and career development.

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Data availability statement

The datasets generated during the current study are available from the corresponding author on reasonable request.

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Ethical approval statement

This study used only secondary data obtained from publicly available sources, including official government websites and publicly accessible databases. The research did not involve human participants, identifiable personal information, or any form of intervention. Therefore, according to standard institutional and international research ethics guidelines, formal ethical approval from an institutional review board or ethics committee was not required.

Informed consent statement

This study did not involve direct participation of individuals or the collection of private or identifiable personal data. All information analyzed in this research was obtained from publicly available sources. Therefore, informed consent from participants was not required in accordance with relevant research ethics regulations.

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Conflict of interest

The authors declare that they have no conflict of interest.

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TABLE 1 Result of test

Hosmer & Lemeshow Test		
Chi-square	df	Sig.
11.799	8	0.16

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TABLE 2 Regression results

Variable	M	SD	B	Sig.	Exp (B)
Control					
Eastern	0.312	0.463	0.696	0.009	2.006
Central	0.265	0.442	0.205	0.372	1.228
Northeaster	0.082	0.274	0.193	0.579	1.213
Sub-provincial	0.087	0.282	-0.979	0	0.376
Growth rate of GDP	0.155	0.154	1.363	0.098	3.907
Gender	0.05	0.226	0.589	0.331	1.802
Ethnicity	0.14	0.351	-0.449	0.132	0.639
Age	47.57	3.902	-0.157	0	0.855
Education	3.86	0.783	0.567	0	1.764
Promotion speed	0.038	0.008	-8.819	0.657	0
Appraisal	0.722	0.448	-0.475	0.045	0.622
Independent					
Party standing	0.618	0.486	0.393	0.038	1.481
Date in office	0.49	0.5	0.505	0.007	1.658
Appointed locally	0.19	0.39	0.565	0.052	1.759
Education gap	0.73	0.73	0.148	0.271	1.159
Diversity gap	1.2	1.05	-0.179	0.032	0.836
Hometown	0.12	0.324	-0.228	0.411	0.796
Alumni	0.12	0.324	-0.296	0.263	0.744
Powerful sector	0.36	0.48	0.229	0.24	1.258

Note: Appraisal refers to whether there is a change in perception regarding the cadre's

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performance. Party standing refers to the longer party standing of the secretary compared to the mayor. Date in office refers to the earlier appointment of the secretary than the mayor. Appointed locally refers to the secretary being appointed locally while the mayor is appointed from other regions. Education gap refers to the absolute value of the education level difference between the secretary and the mayor. Diversity gap refers to the absolute value of the difference in functional experience diversity between the secretary and the mayor. Hometown and alumni refer to the secretary having more relationships with provincial leaders who are from the same hometown or graduate from the same university as the secretary. Powerful sectors refer to the secretary having more experience in powerful departments compared to the mayor.

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